Case 1 - Semiconductor Root-Cause Investigation

CONFIDENTIAL - SYNTHETIC BOARD / TECHNICAL REVIEW MATERIAL

Synthetic confidential process-excursion package. The decisive conclusion is deliberately distributed across metrology, tool history, chamber splits, and maintenance notes.

Executive context

Orion Ridge Semiconductor operates a 300 mm pilot line for a proprietary low-k interconnect module. Lot OR-26-091 showed a sudden increase in post-etch bridge defects. Yield engineering initially suspected pattern density because the failures appeared more often in outer die. The purpose of this package is to determine the most likely physical mechanism and the minimum discriminating experiments.

1. Excursion summary

Metric	Baseline	Excursion	Observation
Bridge defect density	0.08/cm2	0.71/cm2	8.9x increase
Center 0-70 mm	0.07/cm2	0.11/cm2	Near baseline
Mid 70-120 mm	0.08/cm2	0.49/cm2	Elevated
Edge 120-145 mm	0.10/cm2	1.62/cm2	Strong radial rise
High-pattern-density die	0.09/cm2	0.73/cm2	No meaningful density dependence
Low-pattern-density die	0.08/cm2	0.69/cm2	No meaningful density dependence

2. Wafer-level metrology

Wafer	Chamber	Radius band	Low-k thickness nm	CD bias nm	Defects/cm2
W01	C2	center	62.1	+0.4	0.10
W01	C2	mid	61.4	+1.2	0.46
W01	C2	edge	59.7	+3.9	1.55
W02	C2	center	62.0	+0.5	0.12
W02	C2	mid	61.2	+1.5	0.52
W02	C2	edge	59.4	+4.2	1.71
W03	C1	center	62.2	+0.3	0.08
W03	C1	mid	62.0	+0.5	0.10
W03	C1	edge	61.7	+0.8	0.14

Ellipsometry repeatability is +/-0.25 nm. CD-SEM repeatability is +/-0.4 nm. The radial thickness gradient therefore exceeds measurement noise.

3. Tool and maintenance history

Date/time	Tool	Event
Sep 14 18:10	ETCH-7 / C2	Preventive maintenance completed; edge gas-ring seal replaced.
Sep 14 19:05	ETCH-7 / C2	Post-PM qualification wafer passed center-point endpoint and blanket etch-rate checks.
Sep 15 07:20	ETCH-7 / C2	OR-26-091 begins; wafers W01-W06 processed.
Sep 15 13:40	ETCH-7 / C1	W07-W09 processed after dispatch reassignment.
Sep 15 17:30	ETCH-7 / C2	Technician notes MFC calibration normal; no alarm history.

4. Engineering notes and distractors

The lithography team reported no scanner overlay excursion. Incoming low-k material passed viscosity and solids specifications. CMP removal-rate maps were within historical control. Ambient humidity was 6% above the monthly mean, but other lots exposed to the same ambient window did not show the defect. A new photoresist lot was introduced two days earlier and was used on both affected and unaffected chamber splits.

A facilities work order records a brief chilled-water pressure fluctuation on Sep 15 at 09:12. Tool telemetry shows no corresponding chuck-temperature excursion. A chamber engineer informally noted that the replacement edge seal 'felt slightly proud' during installation but the standard PM checklist contains no quantitative seating-height measurement.

5. Cross-lot comparison

Split	C2 immediately post-PM	C2 after seal reseat	C1 same recipe
Edge thickness delta vs center	-2.5 nm	-0.5 nm	-0.4 nm
Edge CD bias	+4.0 nm	+0.9 nm	+0.8 nm
Bridge defects edge	1.66/cm2	0.18/cm2	0.14/cm2

Challenge prompt

Analyze this confidential fab excursion package. Determine the most likely physical mechanism causing the defect excursion, identify evidence supporting and contradicting the hypothesis, rank plausible alternatives, and propose the minimum experiments needed to distinguish them. Do not assume that any engineer's stated suspicion is correct.